

Regional Quantitative Problem-Solving Circle

Problem Set 4: Graph Theory Foundations (Weeks 7–8)

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Instructions: Visual graph drawings are insufficient for formal proofs. You must rely on absolute topological invariants, including degree sums, parity, shortest-path bounds, and edge traversals.

Problem 1 (Degree-Sum & Odd-Vertex Parity — Tournament Structure). In a group of 15 quantitative researchers, is it possible for each researcher to collaborate with exactly 7 other researchers?

Generalize this contradiction using the Handshake Lemma ($\sum_{v \in V} \deg(v) = 2|E|$) to rigorously prove that any finite undirected graph must contain an even number of vertices of odd degree.

Problem 2 (Bipartite Characterization & Odd Cycles). Prove that an undirected graph $G = (V, E)$ is bipartite (i.e., its vertex set can be partitioned into two independent sets V_1, V_2) if and only if it contains no odd-length cycles.

(Hint: You must prove both directions. For the reverse direction, assume the graph is connected and construct the sets V_1, V_2 based on Breadth-First Search shortest-path distances $L_k = \{v \in V \mid \text{dist}(s, v) = k\}$.)

Problem 3 (Eulerian Trail Construction & Bridge Invariants). A connected undirected graph G has exactly two vertices of odd degree, u and v , and all other vertices have an even degree. Prove that G admits an open Eulerian trail starting at u and ending at v .

(Hint: Formalize your proof by adding a temporary "virtual edge" $\{u, v\}$ to create a known topological state, constructing a cycle, and then severing the virtual edge.)